DISCOVER_GRADING_PATH

PREREG: DISCOVER_GRADING_PATH: computationally discovered candidate path predicate, fibre map and ±1 grading

Written 2026-10-24 at about 7:30 PM CDT, after exploration (scripts/explore.py, logs/explore.log; hashes in data/explore.sha256) and before every test below. Owner: Arvin Hampton. Arvin's instruction: "It will need to be computationally discovered." Nothing here is claimed to be a written source. All of it is **Category B** until proved. No bridge claim. Source rule (Architecture_A_Ores_Programme §6): functions on T3 orbits must be labelled as a free-dynamics probe (Option 3). D1–D3 are so labelled.

Exploration findings that motivate the definitions (pre-test, disclosed)

- T3(u) mod 9 is determined by u mod 27.
- The mod-9 charge-transition matrix (over lifts mod 27) has row and column sums 3.
- Hits on class 3 mod 9 never occurred twice in a row in 200 sample paths.

Definitions (exact, re-runnable)

- **Charge of a T3 value:** q(u) = u mod 9. This is the written charge partition C_q evaluated on integers; O_res ⊂ C_3 (x_j = 21+9j ≡ 3 mod 9).
- **Path predicate (Option-3 free-dynamics probe) P_res(u_i)** := [q(u_i) = 3], the "residual-charge hit".
- **Fibre map on hits:** β(u) := ((u − 21)/9) mod 8 for u ≡ 3 (mod 9). It agrees with the written β♯(x_j) = j mod 8 on O_res.
- **±1 grading on O_res (carrier-only, no orbits):** g(x_j) := (−1)^{τ(x_j)}, with τ the written tower label.
- **Paths:** u_0 = SHA3-512(b"holdout-" + str(s)) as an integer, s = 0..999 (holdout, disjoint from exploration seeds); u_{i+1} = T3(u_i), i = 0..538, giving 539 transitions per path.

D1 (exact, Category A once proved; exhaustive check)

For every integer u ≡ 3 (mod 9): T3(u) ≡ 1 (mod 3), so T3(u) ≢ 3 (mod 9). Every predecessor class of charge 3 lies in {0, 4, 5}. Check: all $u < 3 \cdot 10^6$ plus a proof by residues mod 27. PASS iff there are no counterexamples and the proof holds.

D2 (holdout statistics of P_res)

Prediction: over all 1000 × 539 transitions, the hit fraction h satisfies $|h - 1/9| \le 4\sigma$, with $\sigma = \sqrt{(p(1-p)/n)}$, p = 1/9, n = 539 000. Also, zero consecutive hit pairs. PASS iff both hold. **Flag:** a pass is what pseudo-random behaviour predicts; it is not bridge evidence.

D3 (holdout fibre distribution of hits under β)

Prediction: the χ^2 statistic of hit counts over the 8 β-classes vs uniform is ≤ 24.32 (p = 0.001, 7 dof). PASS iff $\chi^2 \le 24.32$. Same flag as D2.

D4 (grading uniqueness on K^+; exact)

Prediction: g is the unique ±1 grading on O_res, up to a global sign, that is (i) constant on every same-tower edge of K^+ and (ii) on path edges (j, j+1) has g(j)g(j+1) = (−1)^{δf(j)}. Check: solve over F_2 on the K^+ 1-skeleton (A2_enrich, read-only import). PASS iff the solution space has dimension exactly 1 over F_2 (affine; the global sign is the kernel) and contains g. **Flag:** the existence half holds by construction. Only uniqueness, which needs connectivity, carries content.

D5 (non-vacuous link test: grading × path)

Prediction: on holdout paths, the hits u with 21 ≤ u < 4880 never occur ($u_i \ge 2^{400}$ for i ≤ 539). So the signed hit sum Σ g over hits that land in O_res is 0 for every path. **Flag:** this is expected from magnitude alone. It documents that P_res cannot reach O_res itself, only its charge class. # RESULTS: DISCOVER_GRADING_PATH PREREG.md sha256 108ab2d4044e21b32e4c60c721e32dfd40f3891d8ab7927321d3321655c1f791, hashed after exploration and before the tests. Scripts: scripts/explore.py (pre-PREREG), scripts/tests.py (hash in scripts/SHA256SUMS). Data: data/tests.json, data/explore.sha256. Logs: logs/. Under 1 s each; nice, OMP_NUM_THREADS=1, ulimit -v 3 GB.

Test	Measured	Verdict	Content?
D1: u ≡ 3 (9) ⇒ T3(u) ≡ 1 (3); predecessors of charge 3 ⊆ {0, 4, 5}	0 counterexamples below $3 \cdot 10^6$; predecessor classes exactly {0, 4, 5}; residue proof (u = 27t + r, r ∈ {3, 12, 21} ⇒ T3(u) = 9t + r/3 ≡ 1 mod 3)	**PASS**, Category A (proved)	exact structure of P_res; not a bridge

Test	Measured	Verdict	Content?
D2: hit fraction 1/9 ± 4σ; no consecutive hits	h = 0.11034 (z = −1.79); 0 consecutive pairs	**PASS**	flagged pre-test as the pseudo-random expectation; the no-consecutive part is forced by D1
D3: hit fibres β uniform ($\chi^2 \le 24.32$)	counts [1933, 14715, 5601, 4, 13053, 14864, 9296, 9]; $\chi^2 = 38633$	**FAIL**, strongly	**genuine discovered structure** (see below)
D4: g = (−1)^τ is the unique ±1 grading (up to sign) constant on towers with path coboundary (−1)^{δf}	consistent, K^+ 1-skeleton connected, solution = g	**PASS**	existence by construction; uniqueness from connectivity. Weak
D5: no hit lands in [21, 4880)	0 such hits (min bit length seen: 137)	**PASS**	expected from magnitude. The PREREG parenthetical “u ≥ 2^400” was wrong (actual min 137 bits); the rule tested was the range condition

Bridge: NO.

Discovered structure (post-hoc explanation of the D3 FAIL; Category A lemma, elementary)

Lemma V (2-adic invariant of T3). - $v_2((4n+2)/3) = 1$ and $v_2((2n+1)/3) = 0$, because 3 is odd. - $v_2(n/3) = v_2(n)$.

Hence, after the first step with $u \not\equiv 0 \pmod 3$, every later value has $v_2(u) \le 1$, i.e. **$u \not\equiv 0 \pmod 4$ for the rest of the path**.

For a hit $u = 21 + 9j$: $u \equiv 0 \pmod 4 \Leftrightarrow j \equiv 3 \pmod 4 \Leftrightarrow \beta(u) \in \{3, 7\}$. **So the fibres β = 3 and β = 7 are forbidden to every hit after the initial pure-division prefix of the path.** The 13 observed exceptions (4 + 9) occur in that prefix. The post-hoc log logs/posthoc_mod4.log shows hits ≡ 0 or 4 (mod 8) only from the n/3 branch (6 cases in 300 paths).

The remaining imbalance (for example class 0 vs classes 1, 5) comes from the same mechanism at higher 2-adic and 3-adic depth. An exact prediction needs a Markov chain on u mod ($2^k \cdot 9 \cdot 8$).

Candidate objects now exact and re-runnable (all Category B as residual ↔ resonant objects; none is a written source)

1. Path predicate P_res(u) = [u ≡ 3 mod 9] (Option-3 free-dynamics probe).
2. Path → fibre map β̃(u) = ((u − 21)/9) mod 8 on hits (agrees with β♯ on O_res).
3. ±1 grading g(x_j) = (−1)^{τ(x_j)} on O_res (carrier-only; unique given D4's constraints).
4. Lemma V: hits avoid β̃ ∈ {3, 7} after the prefix. This is the first non-trivial, exact path → fibre constraint found. # NOTES: DISCOVER_GRADING_PATH ## Pivot On Arvin's correction ("It will need to be computationally discovered") I stopped the written-source hunt. It had produced only rg file lists: no INVENTORY was written, no HUNT_GRADING_PATH folder was created, and no written grading or predicate was confirmed. Earlier negative findings stand: reports/crypto/RPP_spec_v1/RESIDUAL_T3_predicate_sources.md and RPP_formal_statement.md.

Method

1. Exploration (scripts/explore.py, hashed before the PREREG): T3 mod 27 → mod 9 determinism; charge-transition matrix; 200 exploration seeds b"explore-s"; tower-parity grading fibre sums [10, 12, 12, −13, −13, −11, −13, 13] (g total −3).
2. PREREG of D1–D5 with exact definitions; holdout seeds b"holdout-s", s < 1000.
3. Tests; then a post-hoc diagnosis of the D3 FAIL, which led to Lemma V.

Disclosures

- The source rule (Architecture_A_Ores_Programme §6, quoted in RESIDUAL_T3_predicate_sources.md §3) bars carrier functions on T♯ orbits unless they are labelled a free-dynamics probe. P_res and β̃ are so labelled (Option 3). They are **not** residual-carrier definitions and **not** Arvin-written.
- The D5 parenthetical bound was wrong; disclosed in RESULTS.
- The post-hoc mod-8 log reused holdout seeds 0–299; it is diagnostic only.

What still blocks C6/C10/C11/E1–E3

- C6/C10 need a specific 539-step Π word. Discovery gives a predicate family on all T3 paths, not a unique Π. To unblock, Arvin must accept (or reject) P_res/β̃ as "the residual constraint on Π".
- C11 needs a ±1 grading. g = (−1)^τ is now an exact candidate; C11's prediction (68-block sums 0, 67-block defect ±1) can be tested **only after** Arvin accepts g, with a new prereg.
- E1–E3 also need the T3 q-series, which is not discovered here.

Sharpest next step

- New PREREG: an exact Markov chain for T3 on residues u mod (4·9·8 = 288), or deeper, computed by a hashed script, predicting the 8 β̂ hit frequencies. Test on fresh seeds (b”fresh-s”).
- A pass would make the path → fibre map β̂ quantitatively predictable from first principles. It is still residual-internal and still no bridge, but it is the first exact path predicate with content.
- Separately, ask Arvin whether g = (−1)^τ and P_res are acceptable as the grading and predicate before re-running C11/C10.
- order27 and the paper were not touched. CORE_FREEZE; new files only; no push, no sends.